

Verification of Evidence Insufficiency for Cosmological Origin Queries in Restricted Document Sets

Input Question

Where did the Big Bang start?

Domain

Astronomy

Validation Status

needs_data

Problem Statement

The question asks for the spatial origin or starting point of the Big Bang event. However, standard cosmological models posit that the Big Bang was the expansion of space-time itself, not an explosion within pre-existing space. The provided evidence catalog contains no relevant cosmological data, creating a critical knowledge gap.

Rationale

According to the system constraints, only allowed_evidence_ids can be used to establish facts. An analysis of the four allowed evidence IDs (EV-Q048-0ed5d03a917e8ea82986befa, EV-Q048-dc9fb9fe41f8dd3e0b39402c, EV-Q048-004b18a2642627b61a252cb0, EV-Q048-75d7176f83bed02438588186) reveals they pertain to Martian propellant economics, COVID-19 forecasting, NC-FET transistor physics, and neutrino history, respectively. None contain information regarding the Big Bang, cosmic inflation, or the geometry of the early universe. Therefore, no factual claim about the spatial origin of the Big Bang can be supported by the provided evidence.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Insufficient Evidence Hypothesis: The provided evidence catalog contains no data relevant to the spatial origin of the Big Bang, rendering any specific hypothesis about its location or mechanism currently untestable within this system's constraints.
- **Mechanism**: The allowed evidence IDs (EV-Q048-0ed5d03a917e8ea82986befa, EV-Q048-dc9fb9fe41f8dd3e0b39402c, EV-Q048-004b18a2642627b61a252cb0, EV-Q048-75d7176f83bed02438588186) pertain exclusively to Martian propellant economics, COVID-19 forecasting, NC-FET transistor scatter, and neutrino history respectively. None address cosmological spacetime metrics, inflation, or singularity physics required to answer 'Where did the Big Bang start?'.
- **Falsifiable Prediction**: If a comprehensive semantic search of the four allowed evidence cards yields zero mentions of 'Big Bang', 'cosmology', 'spacetime expansion', 'singularity', or 'universe origin', then the hypothesis

of insufficient evidence is confirmed.

- ****Required Observations****: Semantic verification that EV-Q048-0ed5d03a917e8ea82986befa discusses only Mars propellant costs ; Semantic verification that EV-Q048-dc9fb9fe41f8dd3e0b39402c discusses only COVID-19 death forecasting ; Semantic verification that EV-Q048-004b18a2642627b61a252cb0 discusses only NC-FET experimental data ; Semantic verification that EV-Q048-75d7176f83bed02438588186 discusses only neutrino physics history
- ****Risk of Being Wrong****: Low risk regarding current input; however, if hidden metadata or non-textual content in these papers actually contains relevant cosmological data not captured in quoted_text, this assessment would be incorrect.

Hypothesis 2

- ****Hypothesis****: Category Error Hypothesis (Unverified): The question 'Where did the Big Bang start?' presupposes a pre-existing spatial container, whereas valid cosmological models posit that space itself emerged during the event, making 'where' an invalid query parameter without new theoretical evidence.
- ****Mechanism****: Standard cosmology suggests the Big Bang was the origin of spacetime coordinates rather than an explosion within them. Answering 'where' requires evidence distinguishing between expansion-in-space vs. expansion-of-space. No such evidence exists in the allowed set.
- ****Falsifiable Prediction****: If any allowed evidence ID provides observational data supporting a pre-Big Bang spatial framework or a specific coordinate origin for the singularity, this hypothesis is weakened.
- ****Required Observations****: Search for evidence of pre-existing space in allowed IDs ; Search for coordinate-based origin descriptions in allowed IDs ; Verification of spacetime emergence claims in allowed IDs
- ****Risk of Being Wrong****: High risk because this hypothesis relies on general scientific consensus not present in the allowed evidence; it cannot be validated or falsified using only the provided irrelevant sources.

Technical Details

The recommended hypothesis posits that the provided evidence catalog (EV-Q048-0ed5d03a917e8ea82986befa, EV-Q048-dc9fb9fe41f8dd3e0b39402c, EV-Q048-004b18a2642627b61a252cb0, EV-Q048-75d7176f83bed02438588186) contains no data relevant to the spatial origin of the Big Bang. This is a meta-scientific verification task. The experimental design focuses on rigorous semantic exclusion: confirming that each allowed evidence ID pertains exclusively to its stated non-cosmological domain (Martian propellant economics, COVID-19 forecasting, NC-FET transistor physics, and neutrino history respectively). The mechanism involves natural language processing (NLP) based topic classification and keyword absence verification against a controlled cosmological ontology. Since no positive evidence for the Big Bang exists in the allowed set, the 'experiment' is a negative search validation.

Datasets

Source

```json

[

```

{
 "id": "EV-Q048-0ed5d03a917e8ea82986bafa",
 "description": "Text content regarding Martian propellant costs and methalox rocket engines.",
 "relevance": "Irrelevant to cosmology"
},
{
 "id": "EV-Q048-dc9fb9fe41f8dd3e0b39402c",
 "description": "Text content regarding US COVID-19 death forecasting models.",
 "relevance": "Irrelevant to cosmology"
},
{
 "id": "EV-Q048-004b18a2642627b61a252cb0",
 "description": "Text content regarding experimental scatter in Negative Capacitance Field-Effect Transistors (NC-FET).",
 "relevance": "Irrelevant to cosmology"
},
{
 "id": "EV-Q048-75d7176f83bed02438588186",
 "description": "Text content regarding the history and properties of neutrinos in electroweak theory.",
 "relevance": "Irrelevant to cosmology (specifically Big Bang origin location)"
}
]
'''

Target

'''json
{
 "description": "Binary classification output for each evidence ID: 'Cosmology/Big Bang Related' vs 'Not Related'. Expected outcome: All IDs classified as 'Not Related'.",
 "validation_criteria": "Zero false negatives in detecting cosmological content if it were present; zero false positives in identifying non-cosmological content as cosmological."
}

```

'''

## Paper Abstract

Background: The question 'Where did the Big Bang start?' requires evidence from cosmology and general relativity. Method: We analyzed the four allowed evidence IDs provided in the SAGE125 system constraint set. Validation Plan: We propose a semantic keyword search and topic classification experiment to verify the absence of cosmological content in these documents. Results: Pending execution of the semantic verification experiment. The current assessment indicates that all provided sources are topically irrelevant to the query.

## Methods

1. **Semantic Keyword Search**: Execute exact and fuzzy string matching for terms: 'Big Bang', 'cosmology', 'spacetime expansion', 'singularity', 'universe origin', 'inflation', 'metric tensor' within the quoted\_text and metadata of all four allowed Evidence IDs.
2. **Topic Classification Model**: Use a pre-trained scientific text classifier (e.g., SciBERT fine-tuned on arXiv categories) to predict the primary subject area for each document. Verify that predictions align with 'Engineering/Economics', 'Epidemiology', 'Condensed Matter Physics', and 'Particle Physics' respectively, excluding 'Astrophysics/Cosmology'.
3. **Ontology Mapping**: Map extracted entities from each document to a standard scientific ontology (e.g., NASA Astrophysics Data System taxonomy). Confirm absence of mappings to 'Cosmic Origins' or 'Early Universe' branches.

## Experiments

### Baselines

'''json

[

"Random Keyword Match: Baseline probability of finding cosmological terms in random scientific abstracts.",

"General Science Classifier: A broad classifier distinguishing 'Physical Sciences' from 'Life Sciences' to ensure granular topic separation is necessary."

]

'''

### Metrics

'''json

[

"Precision@1 for Cosmology Topic: Must be 0.0 for all four documents.",

"Keyword Hit Rate: Count of cosmological keyterms found per document. Expected: 0.",

"Classification Confidence: Confidence score of the 'Non-Cosmology' label for each document. Expected: >0.95."

]

'''

## Ablation

Remove specific domain keywords (e.g., 'Mars', 'COVID') to test if residual ambiguity triggers false cosmological associations. Test with and without metadata fields.

## Validation Protocol

Cross-validate by manually inspecting the full text snippets provided in the Evidence Catalog. If any snippet contains ambiguous phrasing linking to cosmic origins, flag for deeper review. Otherwise, confirm insufficiency.

## Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

## References

- **\*\*EV-Q048-0ed5d03a917e8ea82986befa\*\*** · arxiv · arXiv:2404.00800
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2404.00800.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:7|section:page-7|paragraph:1;  
content\_sha256=03adabf7a0385bfa42709996f708b86a87e8e385497a2b6166ec2f2325a7c812
- **\*\*EV-Q048-dc9fb9fe41f8dd3e0b39402c\*\*** · arxiv · arXiv:2007.02105
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2007.02105.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:2|section:page-2|paragraph:1;  
content\_sha256=03f8aa647cd75f5ab874d9c89ad9249bdaa8838251f32e35df3f1f877656958c
- **\*\*EV-Q048-004b18a2642627b61a252cb0\*\*** · arxiv · arXiv:1903.03884
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1903.03884.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=0fc49d2917f632d6c8511dfdc2c9217a5da9901f8bddbb60a9c70fa2e592b779

- **\*\*EV-Q048-75d7176f83bed02438588186\*\*** · arxiv · arXiv:hep-ph/9705325
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/hep-ph/9705325.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:2|section:page-2|paragraph:1;  
content\_sha256=136034761e52c5cd5babe265f8efa85d3dfec165bb0626387dc98fe0176fc0b7

## Reviewer Comments

- The candidate hypothesis correctly identifies 'insufficient\_evidence' as the only valid scientific conclusion given the strict constraint to use only allowed\_evidence\_ids.
- All four allowed evidence IDs (EV-Q048-0ed5d03a917e8ea82986befa, EV-Q048-dc9fb9fe41f8dd3e0b39402c, EV-Q048-004b18a2642627b61a252cb0, EV-Q048-75d7176f83bed02438588186) are verified to be topically irrelevant to cosmology or the Big Bang, covering Mars propellant, COVID-19, NC-FETs, and neutrino history respectively.
- The experiment design appropriately treats this as a meta-verification task (negative search validation) rather than fabricating a physical experiment.
- Results field correctly states 'pending' and does not claim execution of the semantic verification.
- No external knowledge, booklet content, or fabricated references were introduced.

## Revision History

## Reproducibility Checklist

- Verify access to full text or sufficient excerpts of EV-Q048-0ed5d03a917e8ea82986befa, EV-Q048-dc9fb9fe41f8dd3e0b39402c, EV-Q048-004b18a2642627b61a252cb0, EV-Q048-75d7176f83bed02438588186.
- Ensure the cosmological keyword list is comprehensive and peer-reviewed.
- Document the version of the topic classification model used.
- Record all negative search results explicitly to prove absence of evidence.